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Interaction of Edge-Cloud Computing Based on SDN and NFV for Next Generation IoT

Zhihan Lv, Wenqun Xiu

Abstract—In order to improve the stability of mobile network system for application of the next generation of Internet of Things, balance the network load and guarantee the quality of user service experience, this paper firstly introduces the computing migration framework for the network of the next generation, and summarizes the concept and content of mobile edge computing using SDN and NFV. And then this paper proceeds to introduce the mobile edge computing strategy based on SDN and NFV technology as well as multi-attribute decision making, computing migration, multi-attribute decision, the mobile edge computing decision model based on SDN and NFV technology and the solving process of the mobile edge computing decision model based on SDN and NFV. Finally, three sets of simulation experiments based on MATLAB are designed to validate the multi-attribute decision of mobile edge computing migration strategy based on SDN and NFV. The results show that the multi-attribute decision making based on SDN and NFV can select the appropriate mobile edge computing center, further reduce the server response time and improve the quality of user service experience. This present study is of great significance to the application of Internet of Things terminal in the next generation of network environment.

Key words—Internet of Things applications; SDN; NFV; Edge computing; Cloud computing

I introduction

A. Background

With the rapid development of Internet of Things applications and smart terminals, according to relevant reports [1], statistics of the aggregate growth of mobile data indicate that the total mobile data is expected to achieve a sevenfold increase in the next four years. Between 2016 and 2019, virtual reality traffic grew from 13.3 petabytes per month to 140 petabytes per month. The proliferation of mobile network traffic poses a huge challenge to existing mobile networks, and the existing network technology no longer meets the business needs arising from the explosion of mobile data [2]. The existing network structure migrates computing tasks to the cloud, which, however, increases the amount of data transmission, brings great pressure to the core network, and undermines the service quality as a result of excessive data transmission.

At present, the research of the next generation network technology goes from strength to strength and research institutions at home and abroad are fully engaged in the research of 5G standard establishment and its key technology [3]. At present, there are relatively few studies conducted on MEC migration strategy, while the main research focus is on the theory and application scenario of mobile edge computing. Alenezi et al. [4] achieved mobile edge computing in mobile network by virtue of SDN, NFV and fog computing. Farshin et al. [5] successfully reduced patient waiting time by means of mobile edge computing in healthcare. Song et al. [6] validated and evaluated the application of mobile edge computing in telemedicine and results showed that mobile edge computing is possible in telemedicine.

MEC is critical in the next generation network technology. By definition, MEC technology is to install the servers with the functions of cloud computing, storage and processing in the accessories of base station, which can provide reliable cloud service for mobile users near the base station [7]. For mobile networks, computing migration strategy can control network traffic, and network load balancing can improve system performance. For users, computing migration strategy can improve the

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quality of user experience service. Therefore, the research of edge computing migration strategy based on MEC is of great practical significance to both users and mobile networks.

B. Introduction to relevant technology

Introduction to the Next Generation Network Environment

From 2G to 3G, from 3G to 4G, the speed of development of mobile network is very fast. When it comes to the next generation network environment, 5G technology undoubtedly will promote the development of Internet of Things, and the reverse holds true. 5G technology can satisfy higher transmission demand, achieve shorter transmission time, and contributes to more efficient network society. The 5G network architecture is divided into system architecture and group network architecture. The logical view of 5G network consists of access plane, forwarding plane and control plane. The network view of 5G network consists of central level, convergence level, regional level and access level [8].

Introduction to SDN and NFV Technology

In the traditional communication network, the network nodes, network topology and hardware are closely coupled. Each service node has specialized equipment, and the device deployment is complex. With the rapid increase and expansion of new network services, the reconstruction and upgrading of node equipment in the traditional network is not only slow, but also with heavy workload, which hinders the development of the new network services. With the development of Software Defined Networks (SDN) and Network Function Virtualization (NFV) technology, its application to the network makes it accomplish new features such as on-demand allocation of resources and business flexibility.

In the original proposed software-defined network architecture, the control plane and the data forwarding plane are separated. The reason is that several controlled forwarding devices constitute the data forwarding plane and the business logic and the forwarding route are controlled by the controller on the control plane controls, thus the software-defined network architecture can realize flexible and convenient control of traffic flow. However, after SDN was proposed, it not only separates the control panel from the data forwarding panel, but also enhances the direct programming control mode, which makes the network better control the data forwarding. The SDN architecture consists of a control plane, an application plane, a data plane, and a control plane [9].

NFV can make general hardware resources map into virtual resources and its main function is to achieve traditional and non-cloud network elements.

C. Introduction to Mobile Edge Computing Based on SDN and NFV

The Concept of Mobile Edge Computing

Mobile Edge Computing (MEC) is a new type of information technology that provides services to mobile users close to mobile wireless access by virtue of cloud computing. NFV, SDN and cloud computing are important technologies in mobile edge computing. These technologies improve the data center server installed next to the wireless base station, which can compress large amounts of data. It is estimated that the data computing center server next to the wireless base station can reduce the use of wired broadband by 36% and the user network transmission time by 49% [10].

The moving edge computing is a form of distributed computing, boasting the characteristics of low latency, fast data processing and intelligence [11]. Mobile Edge Computing does not transfer all data to remote cloud data centers, instead, it filters data uploaded by mobile devices, indicating that data are analyzed and filtered in terms of user layer. This approach not only saves time effectively, but also reduces the pressure on the core network and effectively reduces the data transmitted from the device to the cloud [12-14]. In terms of the intelligentization of mobile edge computing, it transfers the computation and storage from the core network to the edge network, which can effectively reduce network delay. In the meantime, network load, user location and other real-time information can be provided to R & D personnel. During R & D, R & D personnel can add real-time network information to applications and deliver service content to mobile users, which can not only increase the satisfaction of mobile users, but also improve the quality of their service experience.

Research contents of Mobile Edge Computing Based on SDN and NFV

There are six aspects of the main research contents of

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mobile edge computing: (1) The standard protocol is to perform mobile edge computation according to standardized protocol; (2) The development of simulation platform is indispensable in the research of mobile edge computing; (3)Mobile management suggests that specific location of users in the mobile process can be obtained by virtue of mobile management to ensure the quality of service experience of users; (4) Because the pricing model of data flow calculated by moving edge is different from the traditional pricing model, so a new pricing model is needed; (5) Extensibility; (6) Security is the ability of mobile edge computing services to ensure the authentication of visiting users of hosted apps and the security of database.

II Migration Strategy of Mobile Edge Computing Based on Multiple Attribute Decision

A. Migration Decision Model for Mobile Edge Computing

Classic Multiple Attribute Decision Model

In the classical multiple attribute decision model, during the decision making process, not only the set of candidate schemes should be determined, but also the scheme should possess multiple attributes [15]. Given a set X, there are *i* schemes in *X*, denoted as X_1, X_2, …, and X_m. If each candidate scheme has *N* attributes, they are denoted as b_1, b_2, …, and b_n. Because each element in X is discrete, it can be represented as a matrix, as shown in Figure 1. In Figure 1, X_1, X_2, and X_3 represent the three candidate schemes in the decision; b_1, b_2, b_3, b_4, and b_5 represent the attributes of each scheme. Thus in the decision matrix $X_{i.i}$, the attribute b_i is in the candidate scheme X_i.

$$X = \begin{array}{c} \\ X_1 \\ X_2 \\ X_3 \end{array} \begin{array}{c} \begin{array}{ccccc} b_1 & b_2 & b_3 & b_4 & b_5 \end{array} \\ \begin{bmatrix} X_{11} & X_{21} & X_{31} & X_{41} & X_{51} \\ X_{12} & X_{22} & X_{32} & X_{42} & X_{52} \\ X_{13} & X_{23} & X_{33} & X_{43} & X_{53} \end{bmatrix}_{\mathrm{F}} \end{array}$$

igure 1 Matrix representation of classical multi-attribute decision model

Edge Computing Decision Migration Model of Based on SDN and NFV

As is commonly known, the decision index is significantly related to the decision result. Therefore, this paper divides the decision indexes into three aspects: (1) benefit index, which is usually referred to as positive index. The higher the index, the better the result; (2) cost index, also known as reverse index, the smaller the value, the better the result: (3) neutral index, the optimal solution is achieved when the value of the index is taken in the middle of a certain range.

According to mobile edge computing migration scenarios, this paper considers that the mobile edge computing migration is carried out in the mobile network. According to the characteristics of the mobile network, this paper selects five indexes to establish a model, including cost, time, bandwidth, energy consumption and energy calculation to model.

The cost mainly includes two parts: migration cost and communication cost . According to the characteristics of the cost index, the lower is the index value of the model, the better is the scheme design. The equations (1) and (2) represent the migration cost model and the communication cost model, respectively:

$$C_{\mathrm{m}}(si, sj) = W_{\mathrm{m}} * d_{si,sj} \tag{1}$$

where C_{m} represents the cost of migration; $d_{si.si}$ represents the migration cost and the distance of migration ; and W_{m} represents the unit migration cost.

$$C_{\mathrm{c}}(si, sj) = W_{\mathrm{c}} * d_{si,sj} \tag{2}$$

where C_{c} represents the communication energy consumption; $d_{si.si}$ represents the communication distance; and W_{c} represents the unit communication cost.

The time index refers to the time the users need from sending data to receiving data. Time is a cost index, the lower the value, the better the scheme. It consists of two parts, transmission delay and computation delay [16]. The time model is shown in (3):

$$T = T_{\mathrm{p}} + T_{\mathrm{c}} \tag{3}$$

When it comes to bandwidth, it means that there are m virtual machines in a computing center which can serve m users at the same time, and the total amount of bandwidth of a computing center is fixed [17]. Assuming the amount of bandwidth a virtual machine can have is B, the limit values of the number of users in each computing center are set as T1 and T2. Thus the broadband model can be expressed in (4):

$$B = \begin{cases} B_{\max}, m < T_1 \\ W/m, T_1 \le m \le T_2 \\ B_{\min}, n > T_2 \end{cases} \tag{4}$$

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Energy consumption refers to the energy consumed when the MCE computing center offers service to users [18]. Energy consumption is also a cost index. The energy consumption model is shown in (5):

$$E_{\mathrm{si}} = E_{\mathrm{b}} + m * C_{\mathrm{vm}} \tag{5}$$

where E_{si} refers to the total energy consumed when the computer center runs a single virtual machine for m seconds; E_{b} refers to the total standby power consumption of a single virtual machine running in a MEC data center ; and C_{vm} represents the unit energy consumption of the virtual machine.

In this way, the capacity of a computer can be known. The capacity of a computer is fixed and it can serve N users simultaneously, the model is shown in (6):

$$v = \frac{C}{n} \tag{6}$$

where *v* represents the computer capacity acquired by each user; C represents the total capacity of the computer and *n* represents the number of users.

In this paper, cost, time, bandwidth, energy consumption and computation energy are the five factors that influence the multiple attribute decision of MEC computing migration strategy. The experimental decision matrix is shown in Figure 1 (and not explained here). In Figure 1, X_1 , X_2 , and X_3 represent the three candidate schemes in the decision ; b_1, b_2, b_3, b_4, and b_5 represent time delay cost, time, bandwidth, energy consumption and computation energy, respectively. In this paper, the destination of edge data migration is a user-centered circle with the radius of K for data selection, and the set of candidate schemes is composed of these selected data centers.

Because of the large difference between the indexes in the decision-making process, the calculation process is complex, which leads to the inaccuracy of the experimental results. In order to facilitate computation and minimize the difference between the indexes, the indexes should be standardized.

There are two ways of standardization: vector normalization and linear scaling transformation. In vector normalization, index attribute values are scaled down to the same unit length, as shown in (7):

$$y_{i,j} = \frac{x_{i,j}}{\sqrt{\sum_{i=1}^{n} x_{i,j}^{\ 2}}} \tag{7}$$

where $x_{i.i}$ is the original attribute value of the attribute in column J of the i scheme; and $y_{i.i}$ indicates that the attribute value has been normalized. However, each attribute in this method should be normalized.

In linear scaling transformation method, different indexes are standardized differently. In this paper, there are positive indexes (bandwidth, computing power) and negative indexes (energy consumption, time, cost), so the corresponding linear transformation is adopted to standardize the indexes.

(8) and (9) can be used to transform positive indexes:

$$x_j^* = \max_{1<i<m}, x_{i,j} \neq 0 \tag{8}$$

$$y_{ij} = \frac{x_{i,j}}{x_j^*} (1 \le i \le m) \tag{9}$$

where x_j^* represents the minimum value of the attribute in column *j* in the candidate scheme; and y_{ii} represents the standardized attribute value.

And (10) and (11) can be used to transform positive indexes:

$$x_j^* = \min_{1<i<m}, x_{i,j} \neq 0 \tag{10}$$

$$y_{ij} = \frac{x_{i,j}}{x_j^*} (1 \le i \le m) \tag{11}$$

B. Migration Process of Edge Computing Based on SDN and NFV

The mobile edge cloud computing center can detect other mobile edge computing centers within the specified range in the process of computing. Other mobile edge computing centers complete the computing migration tasks through the wired network (such as optical fiber) to ensure the quality of user experience [19, 20]. The computational migration model between mobile edge computing centers is shown in Figure 3.

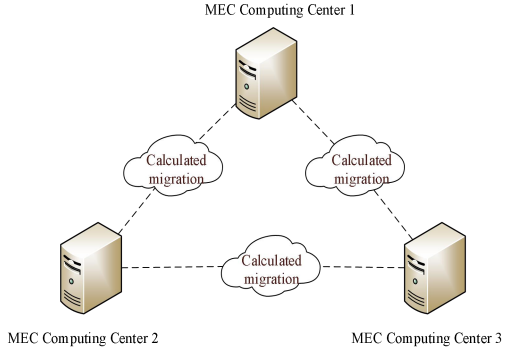


Figure 2 Computational migration between MEC computing centers

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Figure 4 shows the interoperation between MEC and the virtual machine. The source computing center A is the computer center connected by the user mobile terminal and offering service to users. When the user starts to move, the distance between the user and the source computing becomes larger, which results in the increase of transmission time. In order to ensure the quality of user experience, the MEC migration decision based on multiple attribute decision should be launched. Whether the computational migration should be launched or not is decided by the computation results of the multi-attribute decision: if source computing center A is in the first place, no computing migration will be carried out, and the source computing center A will provide services to users; If other computing centers are in the first place, computation is performed along with the computational migration. When the computational migration is finished, the source computing center A will continue to offer services to users.

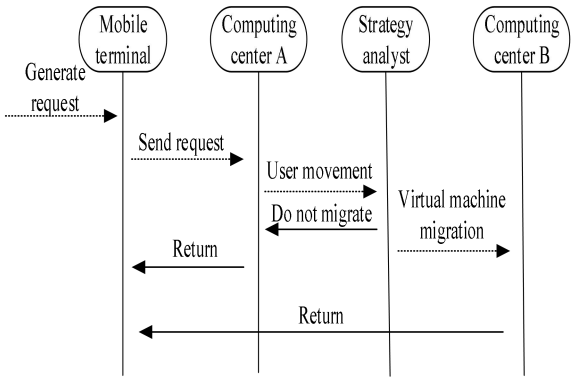


Figure 3 Mobile edge computing virtual machine migration timing diagram

III Simulation and Result Analysis of Edge Computing Migration Strategy Based on SDN and NFV

A. Simulation Design

Simulation scenario

Figure 1 is a system diagram of mobile edge computing migration strategy based on SDN and NFV technology, which consists of initialization module, information gathering module, strategy selection module and evaluation module. The initialization module consists of three parts: initialization of simulation scenarios, initialization of task and initialization of evaluation indexes. In this module, the user's connection to the MEC computing center needs to be demonstrated to initialize computing power, bandwidth, and location. The information collection module mainly records the instantaneous value of the dynamic variables such as the location of mobile devices and the residual computation amount of tasks. The policy selection module is to select and implement the appropriate strategies from NO-migrate strategy, Markov decision and multiple attribute decision. Evaluation index module calculates and records completion time and cost of tasks, energy consumption and bandwidth occupation time.

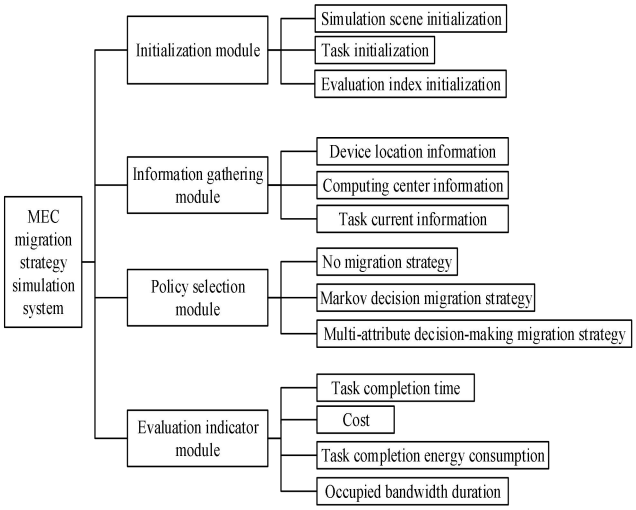


Figure 4 MEC migration strategy simulation flow chart

Simulation parameters

The simulation parameters are shown in Table 1. As can be seen from the table, the parameter settings are: task upload is set as 1M, task calculation is 800 MIPS, server power consumption is [2, 10], task migration is 2M, task downloads are 5M, virtual computing power is [60000, 100000], each server handles unit task costs are [4, 6] * 10 (-4).

Table 1 Simulation parameter design

Meaning	Parameter
Task upload	1M
Task calculation	800MIPS
Server power consumption	[2,10]
Task migration	2M
Task downloads	5M
Virtual computing power	[60000,100000]
Each server handles unit task costs	[4,6] *10(-4)

B. Simulation Results and Analysis

In this paper, three groups of simulation environment are set up. The task types are transmission intensive, computation intensive and IO intensive. For each type of task, the computing center is set under light load, normal

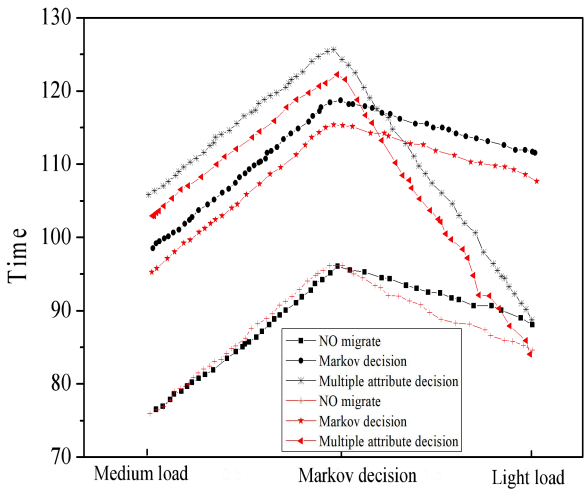
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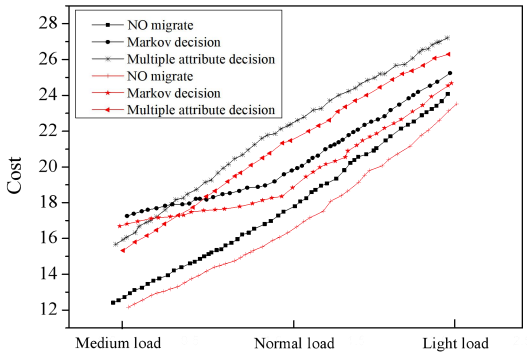
load and heavy load to compare the performance of multiple attribute decision strategy and other two classic strategies. The indices for comparison include task delay, completion cost, network energy consumption and bandwidth occupation time.

Analysis of Simulation Results of Group 1

Figure 5 demonstrates the comparison of simulation performance regarding completion time and cost between the migration strategy proposed in this study and the other two classic strategies under different load conditions in simulation environment 1. In order to ensure the non-contingency of the simulation results, two experiments were carried out for the same strategy and load state. Their simulation results are marked in different colors in Figure 5. It can be seen that the performance of the two experiments is consistent. In terms of task delay, under heavy load and normal load, Markov decision migration strategy and the multiple attribute decision strategy proposed in this study perform better than the non-migration strategy. Similarly, under light load, the multiple attribute decision strategy outperforms the other two. When it comes to task completion cost, experimental results indicate that the performance of multiple attribute decision is weakened under heavy load condition, because in the multiple attribute decision strategy, more weight is dedicated to consideration of time delay performance of the task than the consideration of the distance factor ratio, while the consideration of distance factor is emphasized in Markov decision. Nevertheless, under normal load and light load conditions, the completion cost of the multiple attribute decision is comparable to the Markov decision strategy, which both outperform the NO-migrate strategy.



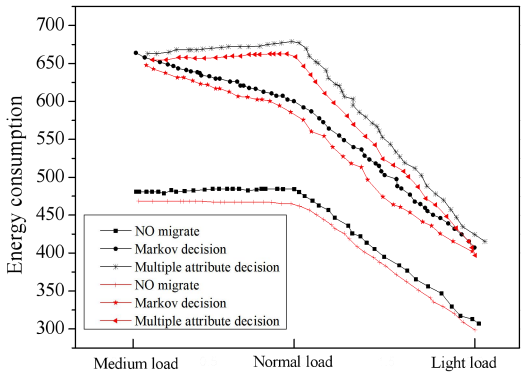
(a) Time



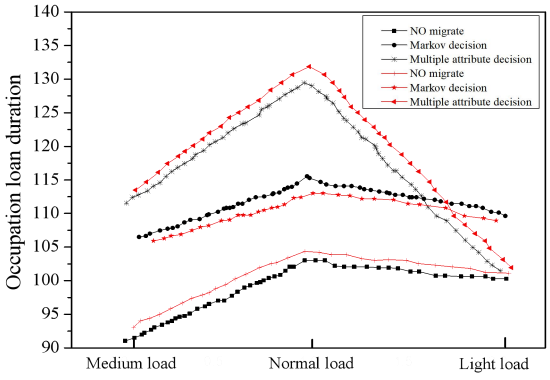
(b) Cost

Figure 5 Completion time and cost of transport-intensive tasks under different migration strategies

Figure 6 demonstrates a comparison of performance regarding energy consumption and bandwidth occupation time among the three strategies. Two experiments under the same experimental conditions both indicate that the multiple attribute decision strategy proposed in this paper considers several factors in the service decision, and optimizes what is neglected by other algorithms. Compared with the other two computing strategies, the multiple attribute decision strategy presented in this paper outperforms them.



(a) Energy consumption



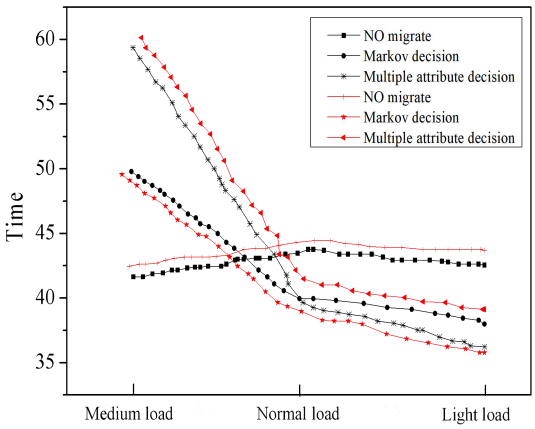
(b)

Figure 6 Energy consumption and bandwidth occupation time of transmission-intensive tasks under different migration strategies

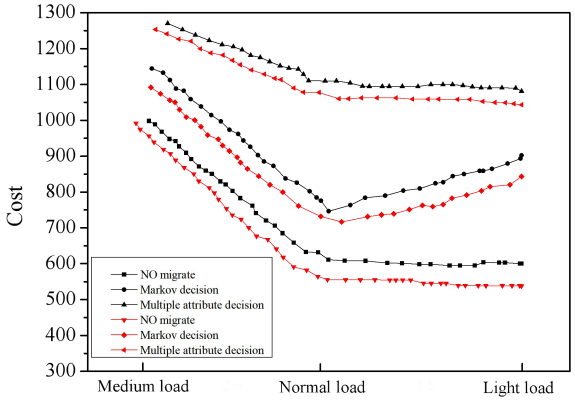
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Figure 7 demonstrates the comparison of simulation performance regarding completion time and cost between the migration strategy proposed in this study and the other two classic strategies under different load conditions in simulation environment 2. In order to ensure the non-contingency of the simulation results, two experiments were carried out for the same strategy and load state. Their simulation results are marked in different colors in Figure 7. It can be seen that the performance of the two experiments is consistent under the same experimental conditions. In addition, the performance of NO-migrate strategy and Markov decision migration strategy demonstrates significant variation in different load states. Among them, the NO-migrate strategy performs well under light load, while Markov decision strategy performs better under heavy load, and both strategies perform poorly in opposite conditions. In contrast, the multiple attribute decision strategy proposed in this paper is more stable in different conditions, can be more widely applied and more advantageous.



(a) Time

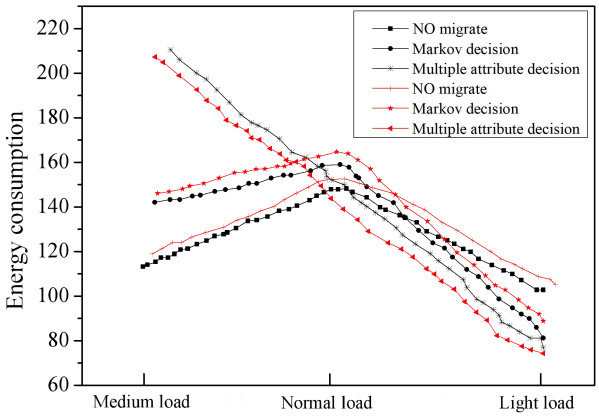


(b) Cost

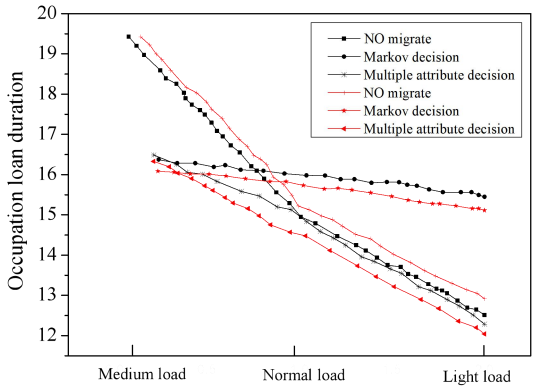
Figure 7 Completion time and cost of computationally intensive tasks under different migration strategies

Analysis of Simulation Results of Group 2

Figure 8 is the comparison of the performance of energy consumption and bandwidth occupation time in simulation environment 2. Two experiments under the same experimental conditions both indicate that the multiple attribute decision strategy presented in this paper is more advantageous than the other two decision strategies and boasts wider possibilities for application. In computationally intensive tasks, the key energy consumption lies in computing and the energy consumption factor has a great influence. The simulation results show that the multiple attribute decision strategy performs better under heavy load and normal load.



(a) Energy consumption



(b) Occupation loan duration

Figure 8 Energy consumption and bandwidth occupation time of computationally intensive tasks under different migration strategies

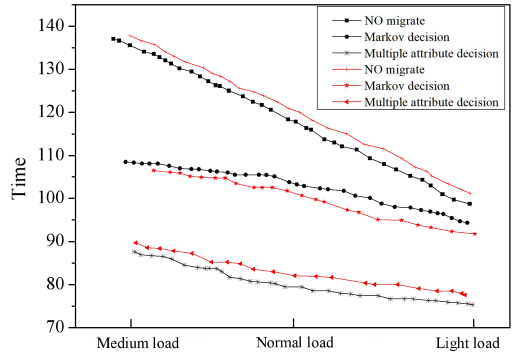
Analysis of Simulation Results of Group 3

Figure 9 shows the comparison of completion time and cost in simulation environment 3. In order to ensure the non-contingency of the simulation results, two experiments are carried out under the same strategy and load conditions. The simulation results are marked in different colors. From the comparison, it can be seen that the performance of the two experiments is consistent. In

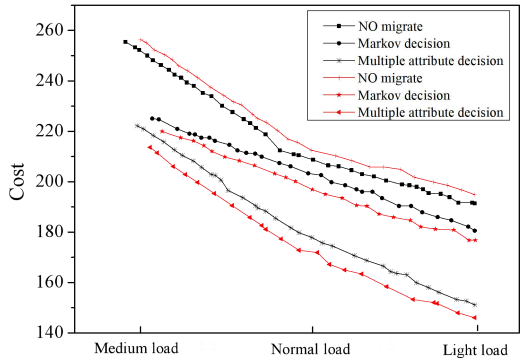
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the IO task, CPU consumption is low, waiting for IO operation occupies the majority of the task. So time is very important and plays a significant role in multifactor computing decision strategy. Therefore, only completion time and cost are compared in this simulation environment. It is shown from the graph that the strategy presented in this paper performs better in completion time and cost under three load conditions, indicating a wide range of adaptability. Results of the two simulation experiments are consistent, which guarantees the reliability of the results.



(a) Time



(b) Cost

Figure 9 Completion time and cost of IO tasks under different migration strategies

Based on the performance comparison of the above three simulation environments, the multiple attribute decision strategy proposed in this paper takes multiple factors into account and performed better in most tasks and load conditions. It is important to design a more reasonable computing strategy to improve network performance.

IV Conclusion

This paper studies the interaction between the edge computing based on SDN and NFV technology and the cloud computing in the next generation Internet of Things. This paper first introduces the concept and implication of mobile edge computing based on SDN and NFV, then illuminates the multi-attribute decision making based on SDN and NFV, and analyzes the solving process of mobile edge computing decision model based on SDN and NFV technology. Finally, three sets of simulation experiments based on Matlab were designed to validate the multi-attribute decision of mobile edge computing migration strategy based on SDN and NFV. In this paper compares the performance results of the three simulation environments, and the results show that considering multiple attributes serves to perform better in most tasks and load status. In addition, two groups of experiments were used in the simulation experiment, which proves that the design of multi-attribute decision computing strategy based on SDN and NFV is more reasonable, and indirectly proves that cloud computing can guide edge computing to serve the network.

This study not only has important research significance for the improvement of network performance, but also promotes the application of Internet of Things terminal in the next generation network environment. In future, we will improve our research by new technology[21][22].

Reference

[1] Muñoz, R., Vilalta, R., Yoshikane N. (2018), Integration of IOT, transport SDN, and edge/cloud computing for dynamic distribution of IOT analytics and efficient use of network resources. Journal of Lightwave Technology, 36(7):1420-1428.

[2] Hsieh H C, Lee C S., Chen J L. (2018), Mobile edge computing platform with container-based virtualization technology for IOT applications. Wireless Personal Communications, 102(4):1-16.

[3] Diro A A., Reda H T., Chilamkurti N. (2018), Differential flow space allocation scheme in SDN based fog computing for IoT applications. Journal of Ambient Intelligence & Humanized Computing, (5):1-11.

[4] Alenezi, M., Almustafa K., Meerja K A. (2018), Cloud based SDN and NFV architectures for IoT infrastructure. Egyptian Informatics Journal, 12(8):32-37

[5] Farshin, A., Sharifian, S. (2017). A chaotic grey wolf controller allocator for software defined mobile network (sdmn) for 5th generation of cloud-based cellular systems (5g). Computer Communications, 108:1231-1238.

[6] Song, S., Lee, C., Cho, H., et.al. (2019). Clustered virtualized network functions resource allocation based on context-aware grouping in 5g edge networks. IEEE Transactions on Mobile

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Computing, (99), 1-2.

[7] Alawe, I., Ksentini, A., Hadjadj-Aoul, Y., and Bertin, P. (2018). Improving traffic forecasting for 5g core network scalability: a machine learning approach. IEEE Network, 32(6), 42-49.

[8] Shah, V. S. (2018). Multi-agent cognitive architecture-enabled iot applications of mobile edge computing. Annals of Telecommunications,73(7-8), 487-497.

[9] Sharma, P. K., Rathore, S., Jeong, Y. S., and Park, J. H. (2018). Softedgenet: sdn based energy-efficient distributed network architecture for edge computing. IEEE Communications Magazine, (99), 2-9.

[10] Kaur, K., Garg, S., Aujla, G. S., Kumar, N. (2018). Edge computing in the industrial internet of things environment: software- defined-networks-based edge-cloud interplay. IEEE Communications Magazine,56(2), 44-51.

[11] Hsieh, H C, Chen, J L, Benslimane M, (2018). Abderrahim. 5G Virtualized Multi-access Edge Computing platform for IoT applications[J]. Journal of Network & Computer Applications, 9(1), 75-79.

[12] Ordonez-Lucena J, Ameigeiras P, Lopez D, et al. (2017). Network Slicing for 5G with SDN/NFV: Concepts, Architectures and Challenges[J]. IEEE Communications Magazine, 55(5):80-87.

[13] Carla M, Jahromi Narjes Tahghigh, Glitho Roch H. (2018). NFV and SDN - based distributed iot gateway for large-scale disaster management[J]. IEEE Internet of Things Journal, 4(9):21-24.

[14] Wei Q, Zhang H. (2017). Future Internet architecture: research status, hot topics, and development practice[J]. Scientia Sinica, 67(2),56-58.

[15] Zarca A M,Bernabe J B, (2019). Trapero, Ruben, et al. Security Management Architecture for NFV/SDN-aware IoT Systems[J]. IEEE Internet of Things Journal, 4(99):1-1.

[16] Pan J, Mcelhannon J. (2017). Future Edge Cloud and Edge Computing for Internet of Things Applications[J]. IEEE Internet of Things Journal, 6(92):109-110.

[17] Gharaibeh A, Salahuddin, Mohammad A, Hussini, Sayed J, et al. (2018). Smart cities: a survey on data management, security and enabling technologies[J]. IEEE Communications Surveys & Tutorials, 9(45), 31-34.

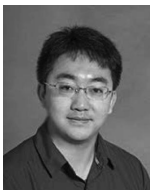
[18] Zhang S L, Laurence T Y, Liwei K, et al. (2019). A Tensor-Based Forensics Framework for Virtualized Network Functions in the Internet of Things: Utilizing Tensor Algebra in Facilitating More Efficient Network Forensic Investigations[J]. 8(3):23-27.

[19] Kaseem A, Bilal R, Dunainawi A. (2018). End-to-End delay enhancement in 6lowpan testbed using programmable network concepts[J]. IEEE Internet of Things Journal, 8(9),12-17.

[20] Farshin A, Sharifian S. (2017). A chaotic grey wolf controller allocator for software defined mobile network (SDMN) for 5th generation of cloud-based cellular systems (5G)[J]. Computer Communications, 108(5), 18-21.

[21] Shi, R., Gan, Y., & Wang, Y. (2018, September). Evaluating scalability bottlenecks by workload extrapolation. In 2018 IEEE 26th International Symposium on Modeling, Analysis, and Simulation of Computer and Telecommunication Systems (MASCOTS) (pp. 333-347). IEEE.

[22] Jin, P., Guo, J., Xiao, Y., Shi, R., Niu, Y., Liu, F., ... & Wang, Y. (2019). PostMan: Rapidly Mitigating Bursty Traffic by Offloading Packet Processing. In 2019 {USENIX} Annual Technical Conference ({USENIX}{ATC} 19).



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